

Zoha Qamar

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EDUCATION

University of Toronto

2025

Honours Bachelor of Science: Computer Science & Statistics

- **Relevant Courses:** Software Design, Theory of Computation, Data Structures and Analysis, Artificial Intelligence, Machine Learning, Databases, Probability and Statistics II, Methods of Data Analysis, Human-Computer Interaction and Communication, Fundamentals of User Interface, UX Design - Prototyping and Evaluation

PROFESSIONAL EXPERIENCE

Application Programmer Co-op

May 2023 - August 2024

Ministry of Education

Toronto, ON

- Developed and tested web applications in an Agile team environment using **VBScript and HTML**, debugging and troubleshooting to improve system reliability and reduce post-deployment issues by **20%**.
- Automated testing workflows and accessibility compliance testing by integrating tools like WAVE, JAWS, and NVDA and ensuring **WCAG 2.0** adherence and enhancing usability for **1,000+ end users**.
- Authored **10+ detailed technical documents** covering workflows, testing protocols, and deployment steps, improving onboarding efficiency by **30%** and streamlining knowledge transfer across development teams.
- Logged, tracked and resolved **40+ software defects** using **ALM and JIRA**, collaborating across cross-functional teams to resolve **95%** pre-release and reduce defect resolution time, maintaining secure, high-quality codebases.

PROJECTS

Personal Portfolio Website with AI Chatbot | *React, Tailwind CSS, React Router, GitHub Pages, Vercel, OpenAI API*

- Designed and developed an interactive, responsive front-end UI pixel-art virtual desk with clickable interactive objects (GitHub, LinkedIn, Resume, About, Contact), attracting 150+ visitors in the first month.
- Built and deployed a Node.js serverless backend on Vercel integrating the OpenAI REST API for real-time chatbot functionality, automating recruiter Q&A, cutting inquiry response time and improving user engagement by 40%.

k-Nearest Neighbors (k-NN) MNIST Classifier | *Python, NumPy, Matplotlib, Pillow*

- Implemented a supervised learning algorithm to classify digits on a subset of 5,000 MNIST images, handling data loading, vectorization, distance computation, and normalization; achieved **90% test accuracy** on the best model.
- Visualized accuracy trends with Matplotlib across varying k values and normalization techniques, identifying optimal hyperparameters and improving model performance by 63% over the unnormalized baseline.

Three Musketeers | *Java, JavaFX, JUnit*

- Collaborated with a group of 3 students to develop a GUI board game using the MVC software design pattern and core OOP principles: abstraction, encapsulation, inheritance, and polymorphism.

Heart Disease Prediction using Decision Trees (NHANES Dataset) | *Python, scikit-learn, Pandas, Matplotlib*

- Built and tuned Decision Tree models to predict heart disease risk on a NHANES dataset of 8,000 patient records, balancing bias-variance trade-offs and achieving **96.8%+ test accuracy** with optimal hyperparameters.
- Conducted EDA and feature engineering by encoding categorical variables, boosting validation accuracy by 25%.
- Visualized feature importance and decision boundaries, enhancing explainability for non-technical stakeholders.

Onitama | *Python, Pygame, Pytest*

- Interactive GUI board game with user-controlled avatars, implementing abstract data types and OOP concepts.

LEADERSHIP EXPERIENCE

Director

July 2022 – April 2023

Women in Science and Computing (WiSC)

- Drove a **30% increase in social media engagement** by launching a strategic content calendar across Instagram, LinkedIn, and Discord, expanding WiSC's visibility to 500+ students.
- Collaborated with marketing and graphic design teams to deliver consistent branding and executed community engagement strategies that boosted event attendance and member retention.

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, JavaScript, C, SQL, PostgreSQL, R, Bash, Shell, VBScript, Assembly

Frameworks: React, Tailwind CSS, NumPy, Pandas, Matplotlib, Scikit-learn, Pytest, JUnit, JavaFX, RESTful APIs

Dev Tools: Git, GitHub, VS Code, PyCharm, Eclipse, Jira, ALM, Linux, Figma, Adobe XD